

Measuring speed improvements

Akamai, a leading CDN provider, has been in the forefront with respect to the adoption of the HTTP/2 standards and has supported the SPDY efforts. A demo hosted by it on <http://http2.akamai.com/> provides the relative timings taken to load its famous tiled Spinning Globe image, both in HTTP/1.1 and HTTP/2. With the former, the latency was 264ms and the load time was 32.7ms. With HTTP/2, the latency was 0ms and the load time was 3.17ms.

CDN provider Cloudflare has the highest number of SPDY and HTTP/2 based servers running across the world, and in its demo site <https://www.cloudflare.com/http2/> the performance improvements are also three times higher when compared to HTTP/1.1 based systems. The time taken to load the tiled image of its server distribution map was 25.18s with the HTTP/1.1 server and 8.34s with the HTTP/2 server.

These improvements are certainly useful in faster loading of websites and real-time updates. This will also increase site affinity with users in locations where broadband connectivity and high speeds are still not available.

Migrating to a new standard

According to a worldwide website tracking software (<https://blog.shodan.io/tracking-http2-0-adoption/>), in December 2015, HTTP 1.1 servers were still three times more than the combined number of various forms of

SPDY and HTTP/2 supported servers. Moving to HTTP/2 standards provides all the advantages that come with the protocol, and a checklist will help in planning and executing the migration.

If the website being migrated already has secure (SSL) support, it will be easy to upgrade the version to the appropriate one that contains this support. If the website doesn't have SSL support, the first step would be to convert it into a secure site (HTTPS) and then ensure that the server has the support for the new standard.

After the server is moved to the latest version, a detailed study of the layout of the resources served from it needs to be made to ensure that the advantages of the concurrency, prioritisation and the push support are appropriately utilised by the applications hosted in the server. Additionally, the SEO optimisations done and image types that are used in applications have to be analysed to ensure that the server's performance can be improved and the functionalities are maintained from the earlier versions. 

By: Maniappan Rajagopalan

The author is a senior enterprise architect with the technology office of the engineering services team in HCL Technologies. He can be reached at maniappan.r@hcl.com.

FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK

We Value Your Feedback

We love to hear from you as **OSFY** consistently strives to make its content **informative, interesting and engaging.**



Please share your feedback/ thoughts/ views via email at osfyedit@osfy.in

We welcome your comments/ suggestions and encourage you to send them to:
The Editor, D-87/1, Okhla Industrial Area, Phase 1, New Delhi-110020.

OpenSource
THE OPEN SOURCE FOR YOU

FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK FEEDBACK